

AMBUJ SHRIVASTAV

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Career Objective

A results-driven Software Development Engineer specializing in architecting scalable, high-availability distributed systems. Eager to solve complex reliability and performance challenges in mission-critical environments like financial technology and large-scale payment processing. Adept at applying deep computer science fundamentals to build robust products that deliver measurable impact at scale.

Projects

SynergyApp - High-Throughput API Gateway:

Spring Boot, JPA, REST, Docker, Bootstrap, Thymeleaf

- Architected and deployed a fault-tolerant, Spring Boot-based distributed system engineered to process over 1,000 requests per minute, directly reducing workflow failures by 50%.
- Executed deep query optimization using strategic indexing, slashing average response latency from 450ms to 95ms and enabling 10x system scalability under load.
- Containerized the application using Docker to create reproducible environments and automated deployment pipelines, reducing manual setup time by 90%.

Logs Guard-AI – Proactive Anomaly Detection:

Scikit-learn, Pandas, Ansible, Docker, Kubernetes

- Engineered an AI anomaly detection system, achieving 85% accuracy and 92% precision on over multiple log entries to proactively prevent system downtime.
- Automated deployment and orchestration using Kubernetes and Ansible, enhancing system resilience and cutting MTTD for anomalies by 40%.

NexPay – Frictionless Payment Framework:

Flutter, Firebase, System Design, API Integration

- Designed the system architecture for a UPI-based payment application, focusing on creating a frictionless, 1-click user payment experience.
- Engineered the business logic for secure transaction processing and user authentication, ensuring data integrity and reliability within the payment workflow.
- Developed a flexible and scalable frontend with a focus on seamless integration with backend payment APIs.

System Enhancement & Utilities

- Developed self-service tools and automation scripts that increased user efficiency by 30% and reduced manual monitoring efforts by 70%.
- Demonstrated proficiency in algorithmic problem-solving and optimization on competitive programming platforms.
- Applied scripting and system-level automation to improve efficiency in real-world utility.

Languages Known

- English
- Hindi

Education

Bachelor of Engineering – Artificial Intelligence & Machine Learning

International School of Business and Media, Pune

Graduation Year: 2024 | CGPA: 8.14

Intermediate (ISC) – 82.0%

Dr. Virendra Swarup Education Centre, Awadhपुरi, Kanpur – 2019

High School (ICSE) – 89.8%

Dr. Virendra Swarup Education Centre, Awadhपुरi, Kanpur – 2017

Technical Skills

- Languages:** Python, Java, JavaScript, C++ (Basics)
- CS Fundamentals:** Data Structures and Algorithms, OOP, System Design, Operating Systems, DBMS, Computer Networks, Complexity Analysis
- Web & Backend:** Spring Boot, FastAPI, Node.js, Express.js, React.js, Next.js, JPA, MongoDB
- Distributed Systems:** Docker, Kubernetes, Ansible, CI/CD, REST APIs
- Machine Learning Tools:** NumPy, Pandas, Scikit-learn, OpenCV
- Problem Solving:** Time & Space Complexity Analysis, Debugging, Algorithm Design, Optimization
- Dev Tools:** VS Code, IntelliJ IDEA, Git, GitHub
- Soft Skills:** Independent Problem Solving, Handling Ambiguity, Team Collaboration, Time Management

Research & Innovation

Research Paper Presented

“AI-Driven Solutions for Driver Safety & Real-time Monitoring”

International Conference on Multidisciplinary Emerging Trends in Engineering and Technology (ICMETET), 2024.

- Proposed scalable AI architectures for public safety systems, addressing real-world deployment challenges in hybrid cloud environments.
- Recognized for effective technical articulation in a global academic forum.

Additional Information

- Maintains an active [GitHub](#) portfolio with projects spanning distributed systems, DevOps automation, and applied AI.
- Strong adaptability and ownership mindset, demonstrated by end-to-end design, coding, deployment, and optimization of real-world projects.
- Organized peer workshops on ML system deployment, mentoring 30+ students